Smart Water Logging, Flood Prevention and Post Flood Event Management System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Deemantha Siriwardana	14/03/2026	

ABSTRACT

Flooding is one of the most common types of natural disaster affecting not only the rural areas but also the urban and vastly developed cities around the world. This causes significant damage to infrastructure, properties and human lives. Many of the flood prone areas around the world lack effective early warning systems and visualizations that allow authorities and residents to understand potential flood risks in advance and accordingly create solutions which will affect greatly in creating solutions to overcome these problems. Existing flood monitoring systems greatly focus on the real time flood data and based on that data provide real time alerts and even so lack predictive flood risk valuations and geographical mapping which help identify vulnerable areas before flooding occurs.

The research gap I have identified is the limited integration of IoT-based systems to gather geographical data and the existing historical data to make future decisions. [1] In this case most current systems rely only on immediate sensor reading taken on the current time which reduces the ability to forecast the areas of impact resulting in damage of property and harm to lives. [2] [3] [4]

The research that I propose is a geo-based Risk mapping system which takes real time geo locational data from IoT Sensors along with predictive results published by the Department of Meteorology [5] and the historical event data of past flood events and predict future flood events and areas which will be affected by them. The IoT Sensors gather information such as water levels and elevation information which provide vital information when developing the flood maps as it helps making the correct elevation data to accurately identify the locations that will be affected. This will help individuals and communities which live and operate in the areas to be affected to evacuate and relocate themselves and even create precautions to the upcoming event of floods which will help immensely in saving lives and properties in the actual time of such events. The flood map generated will showcase areas of effect using colors based on the severity of the effect. The Red Color will indicate the area, which is at the highest risk, the yellow color indicates the areas that are at medium risk, and the green color indicates the areas that are at the lowest risk.

The proposed system will utilize technologies such as IoT sensors and Arduino microcontrollers which will take data readings, Geographical Information Systems to provide areas and geo data of the country, cloud-based data processing and predictive data analytics which collect data to make predictions and visualize them.

The expected outcome of the proposed system is to create a interactive flood risk map which will be generated using the historical data and the predictive data provided by the department of meteorology and the geo data gathered from the IoT devices which are set.

The social and commercial impact would be is that it will create a safer alert system which will provide enough time for users to evacuate from effected areas and even create precautions which will not only save their properties but also their lives. Based on this system flood monitoring agencies and disaster management agencies for such places would be created.

Keywords- Flood Risk Mapping, Internet of Things (IoT), GIS, Flood Prediction, Disaster Management

ACKNOWLEDGEMENT

First of all, I would like to express my sincere gratitude to our project supervisor Deemantha Siriwardhana sir for his valuable guidance and support and constructive feedback to help develop the system and to complete this study.

I would also like to thank my teachers and the staff of the IT department to the supports shown throughout this research.

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TABLE OF CONTENT

Contents

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF FIGURES 6

LIST OF ABBREVIATIONS 6

1. INTRODUCTION 7

1.1 Background and Context 7

1.2 Literature Review 9

1.3 Research Gap 9

2. OBJECTIVE 10

2.1 Main Objective 10

2.2 Specific Objectives 10

3. METHODOLOGY 11

4. HIGH-LEVEL SYSTEM ARCHITECTURE 12

5. USER REQUIREMENTS 14

6. COMMERCIALIZATION PLAN 16

7. BUDGET AND JUSTIFICATION 19

8. WORK BREAKDOWN STRUCTURE (WBS) 20

9. GANTT CHART 21

10. REFERENCES LIST 22

References 22

11. APPENDICES 24

Turnitin Report 25

LIST OF TABLES

Table 1: User Persona 1 16
Table 2: User Persona 2 16
Table 3: User Persona 3 17
Table 4: Cost Table 18

LIST OF FIGURES

Figure 1: UNDRR based on CDRI dataset 7
Figure 2: The share of people living in high-risk flood zones. Source: Rentschler, J, Salhab, M and Jafino, B. 2022. Flood Exposure and Poverty in 188 Countries. Nature Communications. 7
Figure 3: System diagram of Components 13
Figure 4: Work Breakdown Structure 20
Figure 5: Gantt chart 21
Figure 6: React.JS 24
Figure 7: Python 24
Figure 8: JavaScript 24
Figure 9 : GIS technologies 24
Figure 10: Libraries and algorithms 24
Figure 11: UI Mockups 25

LIST OF ABBREVIATIONS

- RP -Research Project
- IoT- Internet of Things
- PDF- Portable document format
- UNDRR - United Nations Office for Disaster Risk Reduction
- SDG- Sustainable Development Goals

1. INTRODUCTION

1.1 Background and Context

Flooding and water logging are two major problems that frequently affect many urban and rural areas of the country. Heavy rainfall and lack of drainage systems or blocked waterways cause the water levels to rise rapidly which leads to floods. In many parts of the world as well as many developing countries there are no systems to predict future flood events which affect many areas facing sudden flash floods which lead to massive property damage and loss of lives.

People in rural communities, as well as people in developed urban areas, business owners and government authorities face numerous issues when faced with these situations as the lack of time to prevent disasters leads to unprepared individuals facing the disaster which leads to loss of life and damage to properties.

According to the report provided by the United Nations office for disaster risk reduction (UNDRR) [6] floods out of the total number of natural disasters take up about 44% globally.

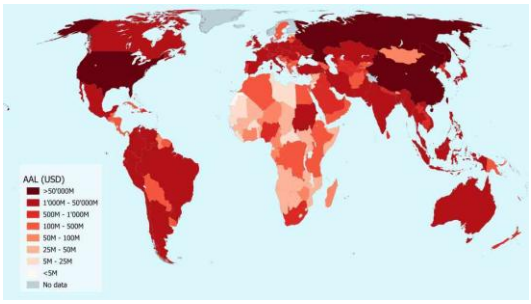


Figure 1: UNDRR based on CDRI dataset

The world bank records proved that around 1.6 billion people are affected by floodings each year. [7]

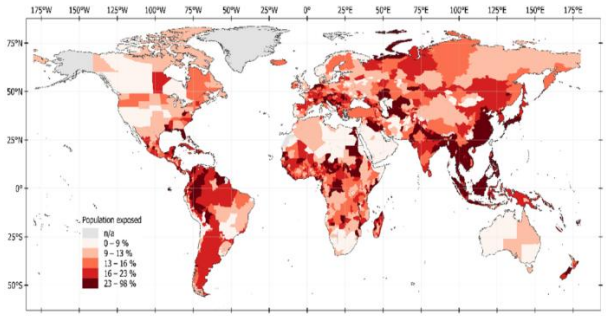


Figure 2: The share of people living in high-risk flood zones. Source: Rentschler, J, Salhab, M and Jafino, B. 2022. Flood Exposure and Poverty in 188 Countries. Nature Communications.

Many researches have been done to address this problem and most of these systems use satellite data, IoT data and water level monitoring systems data to monitor flood scenarios amidst the actual event. These provide insight into the data gathered on sight and this only will help identify flood situation at that time which gives very limited time for people to react and save their lives and properties. Utilizing Remote Sensing and GIS Techniques for Flood Hazard Mapping and Risk Assessment [1] research is a formerly done research which focuses on flood mapping but this takes real time data at sight to generate the map and provide a risk assessment at that time. Researches like Computer Vision and IoT-Based Sensors in Flood Monitoring and Mapping [2], Development of an IoT-Based Flood Monitoring System Integrated with GIS for Lowland Agricultural Areas [3], Development of a Secured IoT-Based Flood Monitoring and Forecasting System Using Genetic-Algorithm-Based Neuro-Fuzzy Network [4] focus on using IoT data to create data visualization of the

time of event and not to predict future events to ensure safety and to help people prepare for such future flood events.

To address these problems my proposed system introduces a flood mapping module as part of the entire smart flood management and post flood event and ration management system. Instead of using real time data as per in many existing systems, my system will collect environmental data using IoT sensers such as water level sensors and rainfall sensors, Then using historical rain fall and flood data as well as the future prediction data given by the department of meteorology creates a predicational flood map which will allow the government authorities, residents and the business owner to know of such events beforehand allowing them to take precautions before the event occur which will save people lives, properties and financial burdens in the face of such events.

The research aligns with few of the United Nations Sustainable Development Goals (SDG) [8] such as 11th SDG which is sustainable cities and communities by contributing to making safer cities by having a well monitored flood prevention and detection system. It also aligns with the SDG 13 which is climate action by supporting communities face a natural disaster caused by extreme rainfalls and floods. It aligns with SDG 9 as it uses many innovative technologies such as IoT devices, data processing and predicting.

The research cluster that the research is done under is the SST or software systems and technologies cluster. By the use of IoT technologies, GIS technologies for geo knowledge gathering and data collection, data analysis and predictive modeling and use of visualization dashboards perfectly align the SST cluster.

1.2 Literature Review

Flood monitoring and flood detection have been studied widely using technologies such as GIS, IoT and machine learning. These have improved flood detection and prevention by providing real time data to users. [1], [2], [3] Some systems use these data to create flood maps to provide visualization of data which help users to ease the search and narrow it down to the exact location and easily identify flood scenarios. [4]

But almost all of this research and system developments have been done using real time data providing less time for users to prepare for such emergencies. All the research data is gathered based on historical events and data gathered in the state of emergency. All of the research has used IoT sensors and geo mappings to gather data.

The use of IoT and GIS in these systems is used only to gain information of the current changes and this will only provide support for a very limited time period, leaving the stakeholders with a very minimal amount of time to focus on providing solutions to this disaster.

1.3 Research Gap

Although much research has been done on the flood detection and prevention systems, existing research and systems only focused on providing solutions and analytical data visualizations in the state of flood [1]. Even though these systems provide valuable information in the state of floods they fail to predict flood risks and risk mappings based on prediction data and historical data gathered by data sources [2]. In most of the existing research single data gathering devices are used and only one area of floods is used to create mapping such as the water levels, flow levels or the rainfall, which is solved by integrated use of many of the data gathering sources to create a unified data resource [3], which can be used to generate the necessary prediction mapping which allows users to gain data a long time before the actual event.

The proposed system will take data from GIS for geo locations of exact places and water levels and flow levels from IoT and Arduino sensors. Using these data and the rainfall prediction data and the historical flood data the system will create a predictive flood zone map few days prior to the exact date of impact allowing users to prepare for such events ahead of time.

2. OBJECTIVE

2.1 Main Objective

Development of a predictive flood risk mapping system which integrates IoT senor data, weather forecast data provided by the department of meteorology and historical data collected by past flood events which will divide the zones based on severity of impact using colors red, yellow and green accordingly.

2.2 Specific Objectives

1. Collect data from the IoT sensors and external data sources such as meteorology department forecasting data sets and the past historical data gathered from former flood incidents.
2. Development of data processing and analysis models which will analyze the data and calculate the level of flood in the areas.
3. Generate a flood risk map using the calculated values and classify areas based on their threat level into high, medium and low flood risk zones using colors red, yellow and green accordingly.
4. To design and implement an interactive geo-visualization interface that displays flood risk zones using colors based on severity.
5. Evaluating performance based on the prediction accuracy, response time and usability.
6. Provide a early disaster preparedness by providing timely and accessible flood risk information prior to the actual disaster to communities and authorities.

3. METHODOLOGY

This research follows the Design Science Research approach as the primary goal is to design, develop and evaluate a flood risk mapping system. This design approach is the most suitable for this research as this study aims to create and test innovative solutions to a real-world problem which is flooding.

This research follows five steps which are problem identification, design and development, implementation, evaluation and refinement. In the problem identification step the problems created by floods and all the limitations of the existing systems are identified and the need for a new system with flood mapping is identified. Design and Development phase and develop the system flood risk mapping system using IoT data and geographic information. In the Implementation step sensors and the data processing modules and the performance of the visualization platforms. Evaluation step is the step where system is tested using environmental data and assessing its predictions accuracy and usability. Refinement is the step where the system is improved based on the problems identified in the evaluation step.

To generate the flood, maps the proposed system will combine the processes of data collection, data processing, predictive analysis and output generation. Data processing step will have collecting environmental data such as water levels, rainfall, water flow rates, weather forecasts and elevation data, filter the data preprocess them and prepare them for analysis. By analyzing this data based on the criteria such as rainfall intensity, river or drainage water levels, terrain elevations and historical flood patterns, the risk level of floods in the area will be identified. Using the analysis data gathered a flood risk map will be generated with GIS technology. The results will be categorized based on high risk, medium risk and low risk. These risk zones will be red, yellow and green colored based on the risk level.

For this development several tools and technologies will be used. IoT sensors such as ultrasonic water level sensors, water flow sensors, rainfall sensors, Esp32 and Arduino UNO 3 micro controller which will gather environmental data and support the flood monitoring process. Programming languages such as Python will be used for the data processing and analysis part whereas JavaScript will be used to develop the system that will showcase the results and to do the actual map building part. For the generation of the flood map Gis Platforms such as QGIs will be used. The sensor data will be stored within a Microsoft SQL database which will ensure that the data is safe and secure.

For this study data will be collected from various different sources such as IoT sensors, Weather forecast data from meteorological services, Geographical elevation datasets and historical flood data records. These records will be used strictly for research purposes of this study.

Ethical considerations:

1. Avoid use of personal and private data
2. Use of publicly available datasets
3. Maintaining data accuracy
4. Ensuring data security

To evaluate the effectiveness of the systems the system will be validated based on several different methods. First method would be testing. The system would undergo unit testing, functionality testing a performance testing and finally usability testing before publication. The performance will be evaluated on several metrices such as prediction accuracy, response time, reliability and availability.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of proposed solution and its components

The proposed system is a predictive flood risk map which integrates IoT environmental monitoring, data processing and GIS technology to identify floods in advance and create a flood map based on the severity of the flood levels. The IoT sensors gather readings such as rain fall, water flow, water level and also predictive rainfall data given by the meteorology department. These data are then fed into a analytical model which analyses these data to then calculate the possible flood values based on the water levels. Using GIS technologies these readings are then turned into a flood map. The flood levels are shown using the colors Red, Yellow and Green based on the level of floods happening to that area. The main objective is to create a predictive flood risk map which enables users to receive flood data at earlier dates making it convenient for them to evacuate and take necessary precautions before the actual flood events.

There are few components in the system.

1. Data collection layer

   In this layer the collection of several types of environmental data required for flood analysis such as Water levels, rainfall level, GED data is collected using many IoT Sensors.

2. Analysis layer

   In this layer the gathered analytical data is processed and using the rain forecast data provided by the department of meteorology is fed into the analytical model. The

   - **NumPy** – numerical calculations
   - **Pandas** – dataset processing
   - **Scikit-learn** – ML models
   - **TensorFlow / PyTorch** – deep learning
   - **Matplotlib / Seaborn** – visualization libraries for the analytical model then processes and analyzes these data using random forest algorithm and sends them to the next layer.

3. Flood risk mapping layer

   Using the data analyzed provided from the next layer and GSI technologies a geographical flood risk map is generated. This map will show the flood risked areas based on the severity in colors red, yellow and green.

4. Database layer

   The Microsoft SQL database stores sensor readings, historical data, rain forecasting data and geographic datasets.

5. User interface layer

   This layer showcases a clean and user-friendly interface which the user can interact and use to gain the information needed about the floods.

4.2 System diagram of components

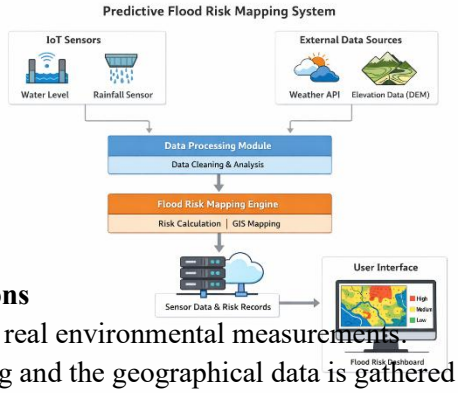


Figure 3: System diagram of Components

4.3 Component interactions

1. IoT sensors collect real environmental measurements.
2. Weather forecasting and the geographical data is gathered from sources.
3. The data processing and analysis model then collects the data from both sources and analyses them.
4. The analyzed data is then used to create a flood risk map using GIS technology.
5. The main database stores sensor readings, historical records and processed results.
6. The user interface displays the generated data and the map in a user-friendly manner.

4.4 Data flow explanation

1. Data gathering- sensors collect environmental data
2. Transmission- Data is cleaned, combined and analyzed
3. Processing- the gathered data is cleaned, combined and analyzed
4. Calculation- the level of flood risk is calculated by an algorithm
5. Mapping- using the flood risk calculated a geographic flood map is developed
6. Visualization- The map is visualized to the users through a web interface

4.5 Scalability

1. Integration of monitoring stations at government offices in the relevant fields.
2. Upscale the areas of monitoring to larger geographical areas.

4.5 Security considerations

1. Secure transmission of sensor data
2. Access control based on roles
3. Data backups to cloud storages on a regular basis

5. USER REQUIREMENTS

5.1 Stakeholder identification

1. Local Residents and communities

Residents who live in or around flood prone areas and communities that are located in and around flood prone areas will use the system to check flood risk levels and the plan the evacuation routes and safe zones prier to actual events of flood.

2. Government authorities

Government authorities like disaster management agencies, flood and disaster management unit and emergency response units will use the system to make plans of recovery in state of such emergency and make plans and funds available to face such upcoming events prior to the actual incident.

3. City planners

Using the data, city planners and the government authorities will make better drainages and plans to avoid such events from occurring.

4. System admins

Admin will update the system, check for errors and reduce down times to ensure the system is available in a state of emergency.

5.2 Functional Requirements

1. Collect environmental data using IoT sensors such as water level sensors, flow sensors and rainfall sensors.
2. Integration of external data sources and weather Api's and databases from the department of meteorology together.
3. Process the data, clean and set the data to analysis
4. The system calculates the level of flood risk based on the forecasted rainfall intensity, geo elevation and the flow rate and impact
5. The flood risk is then mapped using GIS technology based on the severity in colors red, yellow and green.
6. The gathered sensor data, historical data and the forecast data are stored in a database
7. The map and the visualizations are shown in a visually pleasing user interface.

5.3 Non-Functional Requirements

1. **Security:**
    - Ensure the access to all the datasets and data sources are restricted and the ability to modify the system is only available to the administrators.
    - User authentications and role based access control is used to avoid unauthorized access.

2. **Reliability and accuracy:**
    - Ensure the data provided by the data sources are reliable and accurate.

3. **Usability:**
    - The flood risk map should be user friendly and easy to use even by non-technical users.
    - Risk zones should be clearly displayed based on the colors to showcase severity.
    - The application should be accessible using any types of device.

4. **Scalability:**
    - The system should be scalable to support new technologies and locations.

6. COMMERCIALIZATION PLAN

6.1 Target market and customer persona

- Residents and communities of flood prone areas
- Local government authorities
- Disaster management units
- Insurance providers

1) Resident of flood prone area

<table>
<tr><td rowspan="4">  </td><td colspan="2">Noor Shafia</td></tr>
<tr><td>Demographics:
Age: 21
Location: Akurana
Education: Undergraduate
Job Title: Unemployed
Marital Status: Single</td><td>Goals:
Attain flood data and flood locations easily to save their family and their home.</td></tr>
<tr><td>Bio:
An enthusiastic young learner who lives in Akurana. Eager to engage with a new system or technology to reduce the flood damage caused to her family each year.</td><td>Pain Points:
Available systems only provide real time data and with these data ability to avoid these scenarios become impossible.</td></tr>
<tr><td colspan="2">Motivations:
To find a system that will provide flood information prior to incidents and help users avoid incidents and to lessen the damage which will be caused by these incidents by acting fast.</td></tr>
</table>

Table 1: User Persona 1

2) Local government authority

<table>
<tr><td rowspan="4">  </td><td colspan="2">Chamara Perera</td></tr>
<tr><td>Demographics:
Age: 42
Location: Colombo
Education: Master’s degree in urban planning.
Job Title: Municipal Planning Officer.
Marital Status: Married</td><td>Goals:
Access accurate flood risk maps to support urban and city plannings to avoid floods to save cost in post flood recovery and to keep the flow of urban development going.</td></tr>
<tr><td>Bio:
A municipal planning officer responsible for the city development and infrastructure planning. Regularly analyzes environment data and creates adjustments to ensure reduction in floods.</td><td>Pain Points:
Limited access to predictive data. Difficulty identifying future flood risk data due to the lack of analysis of former data.</td></tr>
<tr><td colspan="2">Motivations:
To utilize flood maps to design safer urban environments and improve drains to prevent future floods and ensure safety of communities.</td></tr>
</table>

Table 2: User Persona 2

3) Disaster management units

<table>
<tr><td rowspan="4">  </td><td colspan="2">Nadeesha Fernando</td></tr>
<tr><td>Demographics:
Age: 36
Location: Kandy
Education: Bachelor’s Degree in Disaster Management
Job Title: Emergency Response Coordinator
Marital Status: Married</td><td>Goals:
Monitor Flood risks prior to incident and stock supplies and make preparation to face such scenarios prior to the incident.</td></tr>
<tr><td>Bio:
A junior emergency response coordinator working in the disaster management unit of the Kandy municipal council. She is responsible for monitoring disaster risks and planning evacuation and rescue plans during disasters.</td><td>Pain Points:
Lack of a flood prediction system that will accurately predict data to prepare in the face of a flood.
Difficulty planning evacuations as flood information arrive too late.</td></tr>
<tr><td colspan="2">Motivations:
To use predictive flood monitoring system that enable early warnings to make early decision and pre prepare for future events and stock supplies.</td></tr>
</table>

Table 3: User Persona 3

4) Insurance Providers

<table>
<tr><td rowspan="4">  </td><td colspan="2">Nadeesha Fernando</td></tr>
<tr><td>Demographics:
Age: 45
Location: Kaduwela
Education: Bachelor’s degree in finance
Job Title: Insurance plan creation and risk assessment officer
Marital Status: Married</td><td>Goals:
Asses area which faces floods and introduce plans to insure properties and to those areas.</td></tr>
<tr><td>Bio:
A Insurance plan creation and risk assessment officer working at Srilanka insurance company for 10 years. His role is to evaluate areas based on risk and create insurance plans suitable for those geographical locations.</td><td>Pain Points:
Difficulty obtaining correct and accurate flood risk data for specific locations.
High financial loss due to unavailability of insurance plans in places facing floods.</td></tr>
<tr><td colspan="2">Motivations:
To use predictive flood monitoring system to make plans prior to incidents and to improve insurance risk evaluations prior to incidents making it more effective.</td></tr>
</table>

6.2 Value proposition

- **Predictive flood mapping:** Comparing to all the other available systems which follow traditional data gathering done in face of disaster the proposed system uses forecast data and real time sensor data to provide accurate predictive mapping of flood risks.
- **Early warning support:** Due to availability of flood predictions the system provides early warnings to such scenarios making it possible for user to prepare and avoid consequences caused by such events.
- **Real time updates:** Data is constantly monitored to ensure data stays accurate at all times.
- **Data- Driven Decision making:** Provides visual insights that support disaster management planning and resource allocation much easier.

6.3 Revenue model

1. Government license model

For disaster management and recovery operations government authorities can buy a special license.

2. Data analytical service

Environmental authorities and research can access gathered data and historical data and previous analysis reports via paid access management.

3. Custom Deployment Services

Based on user requirements the system can be changed to display different information for a implementation fee.

6.4 Cost estimation

	Unit price	Units	Total price
Arduino microcontroller	Rs.3500	3	Rs.10 500
Water level sensor	Rs.2400	6	Rs. 14 400
Rainfall sensor	Rs.1800	6	Rs. 10 800
Communication module	Rs.1500	2	Rs. 3 000
Cloud host (per month)	Rs. 2500	8 months	Rs. 20 000
Miscellaneous	Rs.2000	8	Rs. 16 000
WIFI/broadband connection (8 months)	Rs.2500	8 months	Rs. 20 000
		Total	Rs. 94 700

Table 4: Cost Table

7. BUDGET AND JUSTIFICATION

	Justification	Type of expense	Unit price	Units	Total price
Arduino microcontroller	Used to connect the sensors together	Hardware	Rs.3500	3	Rs.10 500
Water level sensor	To gather data on the water levels of different parts of the city	Hardware	Rs.2400	6	Rs. 14 400
Rainfall sensor	Measuring rainfall intensity	Hardware	Rs.1800	6	Rs. 10 800
Communication module	Centralized	Hardware	Rs.1500	2	Rs. 3 000
Cloud host (per month)		Licenses	Rs. 2500	8 months	Rs. 20 000
Miscellaneous		Hardware/other	Rs.2000	8	Rs. 16 000
WIFI/broadband connection (8 months)			Rs.2500	8 months	Rs. 20 000
				Total	Rs. 94 700

8. WORK BREAKDOWN STRUCTURE (WBS)

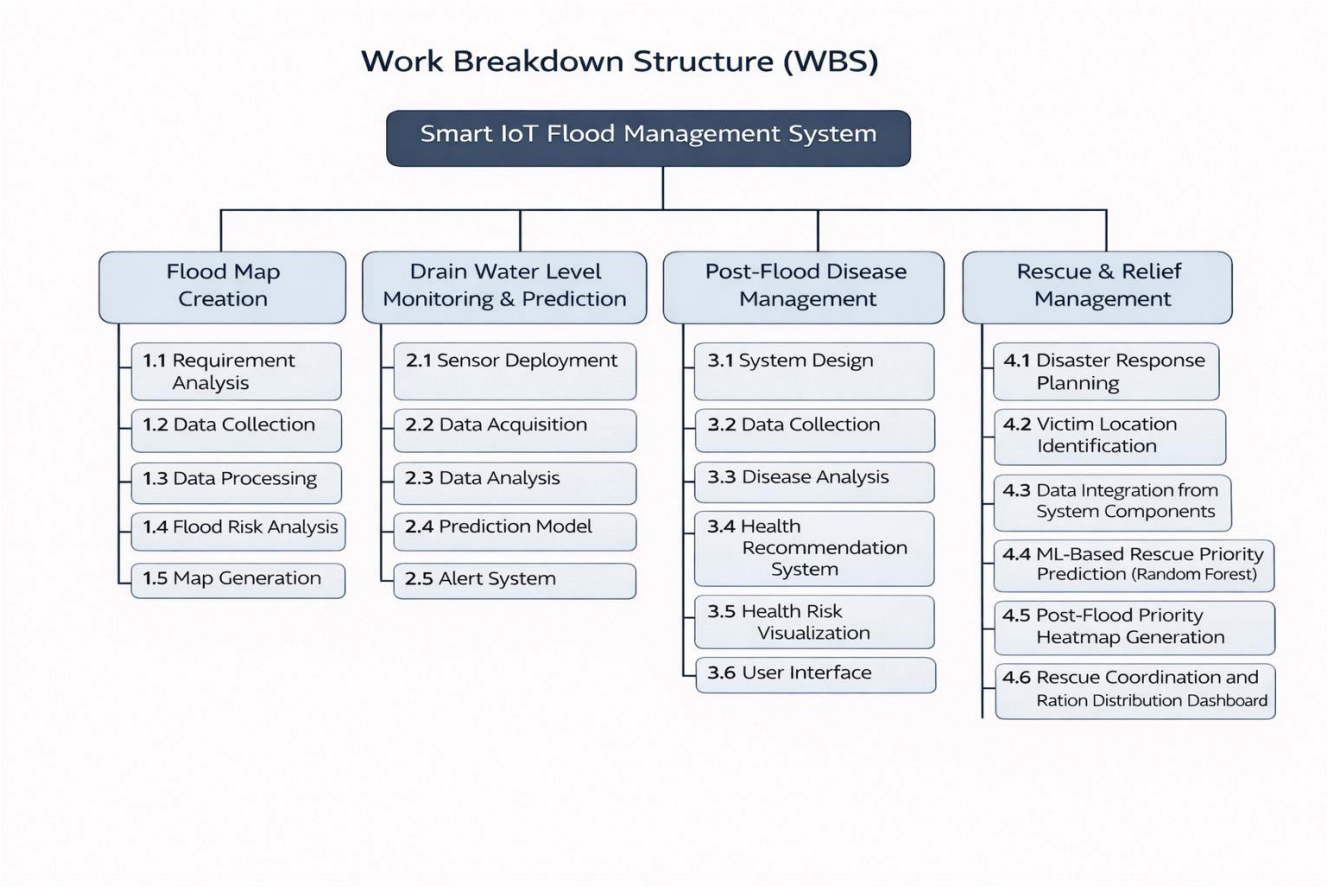


Figure 4: Work Breakdown Structure

9. GANTT CHART

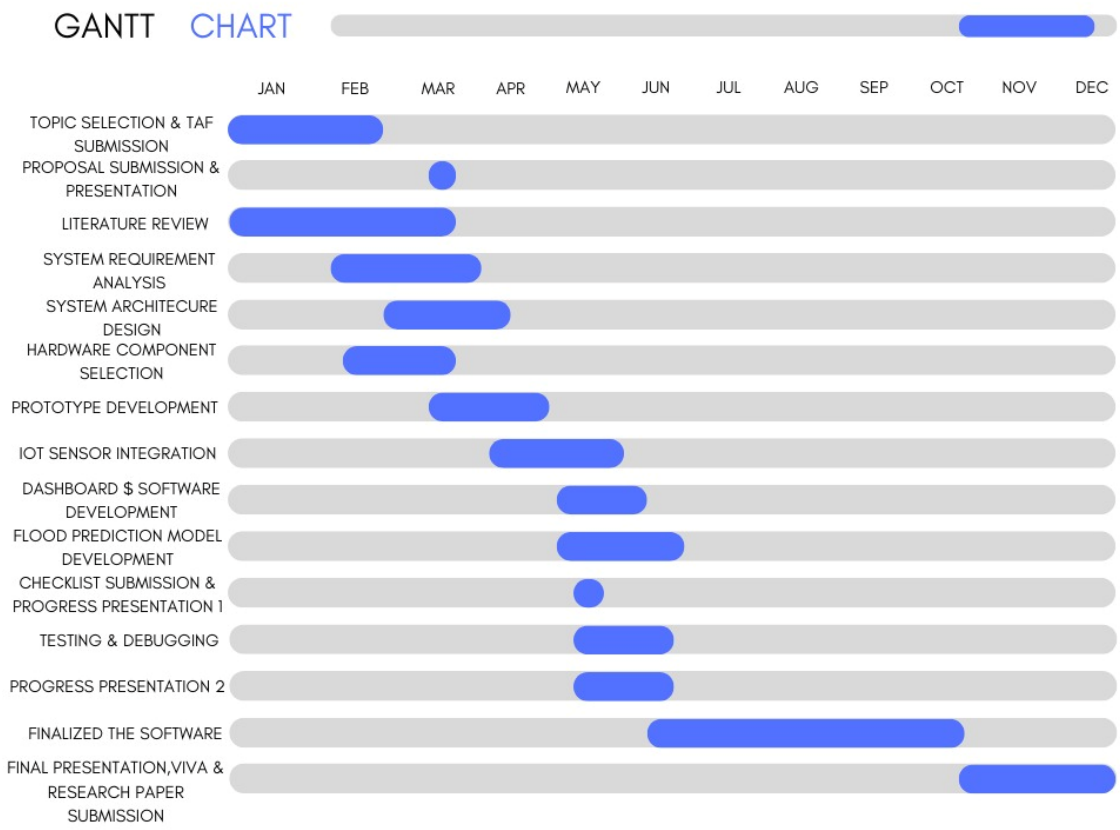


Figure 5: Gantt chart

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11.APPENDICES



Figure 8: JavaScript

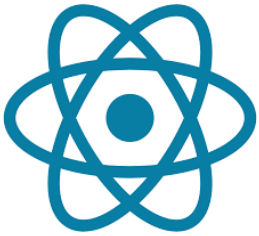


Figure 6: React.JS



Figure 7: Python



Figure 9 : GIS technologies



Figure 10: Libraries and algorithms

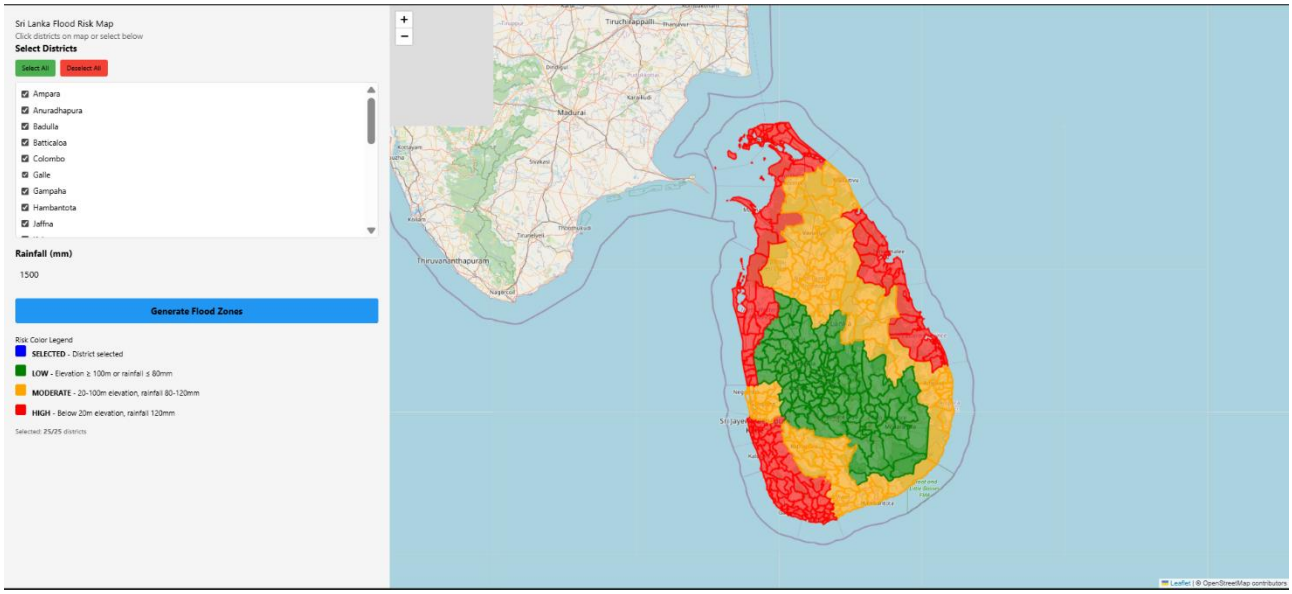


Figure 11: UI Mockups

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